

FORT SHEVCHENKO, Kazakhstan

WMO: 380010

Lat: 44.5476N

Lon: 50.2425E

Elev: -82

StdP: 14.74

Time Zone: 5.00 (E05)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.9	11.8	-3.6	4.5	9.0	1.7	6.0	15.9	34.7	31.2	31.4	30.9	11.3	50	0.622

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.7	98.9	66.1	96.5	65.7	93.0	65.6	78.2	84.7	76.0	82.8	73.9	81.4	13.1	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.3	136.9	82.8	73.8	125.4	80.9	71.4	115.6	79.3	41.7	84.5	39.4	82.5	37.5	81.4	86.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
31.7	27.0	23.0	0.7	104.2	9.0	2.5	-5.7	106.0	-11.0	107.5	-16.1	108.9	-22.6	110.7
			1.3	80.4	9.6	3.0	-5.6	82.5	-11.2	84.3	-16.6	86.0	-23.6	88.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	55.0	29.9	32.3	41.7	53.3	65.7	75.6	80.8	79.4	68.5	55.3	42.6	33.7	(6)
(7)		DBStd	19.50	8.81	8.80	6.70	7.17	7.37	6.58	5.36	6.12	6.82	7.61	8.05	7.70	(7)
(8)		HDD50	2214	624	496	269	43	0	0	0	0	36	240	506		(8)
(9)		HDD65	5161	1089	916	723	357	83	4	0	1	39	308	671	970	(9)
(10)		CDD50	4050	0	0	11	141	487	768	956	912	554	202	19	0	(10)
(11)		CDD65	1524	0	0	0	5	105	322	491	448	144	9	0	0	(11)
(12)		CDH74	16554	0	0	1	48	914	3412	5777	5161	1173	68	0	0	(12)
(13)		CDH80	7657	0	0	0	7	316	1597	2809	2516	405	7	0	0	(13)
(14)	Wind	WSAvg	11.5	11.9	12.2	12.5	12.2	10.8	11.4	10.5	10.3	11.0	11.2	11.5	12.3	(14)
(15)	Precipitation	PrecAvg	5.6	0.2	0.3	0.5	0.6	0.6	0.5	0.4	0.3	0.5	0.4	0.6	0.5	(15)
(16)		PrecMax	13.3	0.9	1.1	1.7	2.6	4.4	3.4	2.0	2.2	3.0	1.2	2.0	1.4	(16)
(17)		PrecMin	2.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	2.5	0.2	0.3	0.5	0.7	0.9	0.7	0.5	0.5	0.6	0.3	0.5	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.1	54.5	66.0	79.2	93.3	100.4	102.4	102.6	94.5	80.3	64.8	53.4	(19)
(20)		MCWB	44.8	47.3	51.0	58.1	62.0	65.5	67.7	66.8	65.2	60.4	52.6	47.1	(20)	
(21)		2%	DB	45.0	48.7	59.3	73.5	87.8	96.6	99.0	98.8	89.1	75.3	59.8	48.8	(21)
(22)		MCWB	42.4	43.1	47.7	56.2	60.7	64.2	66.4	66.4	61.7	57.6	52.7	44.3	(22)	
(23)		5%	DB	42.8	45.8	55.2	69.3	82.8	93.1	96.4	95.4	84.4	71.4	57.2	46.3	(23)
(24)		MCWB	40.2	41.5	46.3	55.3	60.5	64.0	66.3	65.9	61.1	57.3	51.5	42.7	(24)	
(25)		10%	DB	40.6	43.0	51.5	64.7	78.8	89.4	92.8	91.7	80.6	67.6	54.0	43.6	(25)
(26)		MCWB	38.3	39.6	44.6	53.4	60.4	64.0	66.9	66.2	61.1	56.8	49.3	40.5	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.7	48.0	52.8	62.7	69.6	76.7	81.6	82.3	73.8	67.5	57.4	48.6	(27)
(28)		MCDWB	48.0	53.3	61.7	70.6	78.2	82.4	87.6	88.8	80.2	72.6	61.0	51.5	(28)	
(29)		2%	WB	42.7	44.3	49.9	59.1	66.7	73.9	78.5	78.6	70.6	63.6	54.6	45.3	(29)
(30)		MCDWB	44.8	47.5	57.0	68.3	75.3	80.2	84.9	85.3	77.9	69.0	57.9	48.0	(30)	
(31)		5%	WB	40.3	42.1	47.5	57.0	64.9	71.9	76.5	76.0	68.4	60.9	52.3	43.0	(31)
(32)		MCDWB	42.2	44.9	53.3	65.1	73.8	79.1	83.0	83.2	75.7	66.7	56.0	45.3	(32)	
(33)		10%	WB	38.1	39.9	45.4	55.1	63.3	70.1	74.5	73.6	66.4	58.7	49.8	40.7	(33)
(34)		MCDWB	39.9	42.3	49.9	62.7	72.7	78.2	81.9	81.6	74.2	64.3	54.0	42.9	(34)	
(35)	Mean Daily Temperature Range		MDBR	12.6	14.2	15.1	17.3	18.6	20.2	19.7	20.9	19.8	18.0	14.6	12.3	(35)
(36)		5% DB	MCDBR	13.0	17.3	21.9	24.4	25.9	26.5	26.0	27.3	25.7	24.1	18.9	15.0	(36)
(37)		MCWBR	11.1	13.1	13.1	12.8	12.6	12.9	12.8	13.9	14.6	15.0	13.9	12.1	(37)	
(38)		5% WB	MCDBR	12.7	15.3	19.4	20.1	20.3	19.3	17.9	19.2	19.6	17.8	16.7	13.2	(38)
(39)		MCWBR	11.1	12.5	12.9	11.9	12.0	12.8	12.9	13.7	14.9	14.3	13.5	11.7	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.393	0.413	0.448	0.517	0.497	0.483	0.492	0.491	0.458	0.422	0.399	0.405	(40)	
(41)		taud	2.107	2.072	2.037	1.900	2.020	2.109	2.105	2.082	2.112	2.167	2.142	2.069	(41)	
(42)		Ebn at Noon	214	235	245	238	249	252	248	243	239	230	210	193	(42)	
(43)		Edh at Noon	32	40	47	58	54	49	49	48	43	36	31	31	(43)	
(44)	All-Sky Solar Radiation	RadAvg	398	703	1114	1586	2002	2239	2135	1921	1480	891	495	292	(44)	
(45)		RadStd	41	89	109	147	138	109	109	80	99	97	55	39	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	N/A	-3.01	-3.87	N/A	N/A	N/A	N/A	
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page